

# RAVITEJA MADDELA

MSc Computer Science & Engineering Student | Ex-Software Developer (6+ Yrs Exp)

 Copenhagen, Denmark  +45 71806856  s252744@student.dtu.dk  tejamaddela9@gmail.com  github.com/teja-mr-1  
 linkedin.com/in/raviteja-m-33a413263

## PROFESSIONAL SUMMARY

MSc Computer Science student at the Technical University of Denmark (DTU) with 6+ years of industry experience as a Software Developer across healthcare, fintech, and banking domains (GE Healthcare, Swiss Re, American Express). Specializing in Machine Learning systems, Multi-Agent pathfinding, web applications, and scalable backend infrastructure using PyTorch, Java, and Python.

## EDUCATION

**Technical University of Denmark (DTU)** 2025 – Present  
**Master of Science in Computer Science & Engineering** (Copenhagen, Denmark)  
Key Focus: Machine Learning Systems, Multi-Agent Systems, Algorithm Engineering, Distributed Systems. Student ID: s252744.  
**RVR & JC College of Engineering (Acharya Nagarjuna University)** 2012 – 2016  
**Bachelor of Technology in Information Technology** (India)

## TECHNICAL SKILLS

**AI & ML Systems:** PyTorch, QLoRA, Transformers, RAG, Mixture-of-Experts (MoE), BioClinicalBERT, FAISS  
**Backend & Web:** Python, Java, JavaScript, Leaflet.js, Mapbox, FastAPI, Spring Boot, REST APIs, React, Tailwind CSS  
**Cloud & Tools:** Docker, Git, Linux, AWS, Kaggle, HuggingFace, Netlify

## FEATURED ENGINEERING & RESEARCH PROJECTS

**Sparse MoE-QLoRA for Danish Legal QA** PyTorch MoE QLoRA Personal Research Project  
• Engineered a 4-bit Quantized Low-Rank Adaptation (QLoRA) pipeline to fine-tune a 7B LLM for Danish legal domain QA.  
• Implemented centroid-based Sparse Mixture-of-Experts (MoE) routing using E5 embeddings paired with FAISS RAG retrieval.  
• Achieved 6.88 tokens/sec generation throughput and sub-5ms in-place VRAM adapter hot-swapping on a single Kaggle T4 GPU.

**Weather-Aware Route Planner & Travel Advisor** JavaScript Leaflet.js Mapbox Personal Project  
• Built an interactive route planning application combining Leaflet.js and Mapbox GL JS map rendering.  
• Integrated OpenWeatherMap APIs to sample weather conditions along route waypoints in real time.  
• Implemented AI travel recommendations for trip safety and Netlify proxy functions for secure API key handling.

**Multi-Agent Coordination in Hospital Domain** Java CBS A\* Search DTU 02285 Course Project  
• Developed a Java multi-agent path finding (MAPF) solver for autonomous robot agents in the MAVis Hospital Domain grid world.  
• Implemented Hungarian matching for box-goal allocation, fast greedy priority orderings, and 3-phase box staging.  
• Built a Conflict-Based Search (CBS) repair engine with CAT bias, opportunistic bypass, Lemma-1 jump points, and segment replanning.

**Multi-Modal Clinical Phenotyping System** PyTorch FastAPI FHIR R4 Healthcare AI Project  
• Built a dual-encoder PyTorch pipeline pairing BioClinicalBERT for unstructured clinical notes with a custom Masked Lab-Test Transformer (MLTM) for numerical lab features on MIMIC-IV data (3,397 patient records, 25 conditions, 0.88 AUROC).  
• Developed a FastAPI inference service with SHAP lab importance, BERT attention scores, FHIR R4 bundle adapters, and Gradio web interface.

## PROFESSIONAL EXPERIENCE

**Software Developer** Jan 2023 – Dec 2023  
**Infinite Computer Solutions (Client: GE Healthcare)** (Bengaluru, India)  
• Engineered internal license management software for hospital equipment (MRI and CT scanners), enabling staff to easily activate, update, and manage feature licenses without manual intervention.  
• Streamlined automated installation and upgrade tooling, reducing system downtime and manual overhead for active hospital scanners.

## Software Developer

Feb 2019 – Jul 2022

**Tech Mahindra (Client: Swiss Re)** (Hyderabad, India / Zurich, Switzerland)

- Developed *SwiftMX*, a transaction conversion tool mapping insurance business team requests into automated bond and security purchase orders via third-party financial tools (Apex).
- Built testing and data migration utilities for TRAX application upgrades, verifying config transfer success rates to guarantee zero downtime.
- Implemented custom encryption middleware routing TRAX payment files (salary and insurance disbursements) through multi-system networks (Swift) securely to banking institutions.

## Junior Software Engineer

Sep 2016 – Jan 2019

**Tata Consultancy Services (Client: American Express)** (Hyderabad, India)

- Developed online PDF credit statement generation features for American Express Mexico internet banking, replacing inaccessible XML formats.

## CERTIFICATIONS & LANGUAGES

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**Certifications:** Python Institute Certified Programmer | Lean Six Sigma Yellow Belt | TOEFL iBT (Score: 94/120)

**Languages:** English (Fluent), Telugu (Native), Danish (Prøve i Dansk 2 / PD2)